pharynx and larynx producing chocking or gagging that patients develop a dread of even the sight or sound of water (hydrophobia).

Animal Infection

In dogs, the incubation period is **usually 3–6 weeks** but it may range from 10 days to a year. The initial signs are an **alert, troubled air and restlessness, snapping at imaginary objects, licking** or gnawing at the site of the bite.

After 2–3 days of this prodromal stage, the disease develops into either the furious or dumb types of rabies. In **furious rabies**, dog runs biting without provocation, the lower jaw droops and saliva drools from the mouth. Paralysis convulsions and death follow. **In dumb rabies,** is the paralytic form, animals lies huddled, unable to feed. About 60% of rabid dogs shed the virus in saliva. Rabid dogs usually die in 3-5days.

Laboratory Diagnosis

Human Rabies

The specimens tested are **corneal smears and skin biopsy**. Commonly used method for diagnosis is the demonstration of **rabies virus antigens** by immuno fluorescence. **Direct immunofluorescence** is done using antirabies serum tagged with fluorescein isothiocyanate.

Negri bodies in the brain, are demonstrated, Isolation of the virus by intracerebral inoculation in **mice** can be attempted from the brain, CSF, saliva and urine. The mice are examined for **signs of illness**, and their brains are examined at dealth.

Animal Rabies

The **whole carcass** of the animal suspended to have died of rabies may be sent to the laboratory. The brain may be **removed** sent for biological test and microscopy respectively. The portion of brain sent should include the **hippocampus and cerebellum** as **negri bodies** are most abundant. The following tests are done in the laboratory.

1. Demonstration of rabies virus antigen by immuno fluorescence
2. Demonstration of inclusion bodies - Negri bodies are seen as intracytoplasmic, round or oval purplish pink with characteristic basophilic inner granules. Negribodies vary in size from 3.27 Mm.

Infobits

Local Treatment for rabies

- Prompte cauterization of the wounds helps to destroy the virus.
- Antirabic serum may be applied topically.
- Antitetanus measures and antibiotics to prevent sepsis.

Antirabic Vaccines

Antirabic vaccines fall into two main categories neural and non-neural.

Neural Vaccines

Suspension's of nervous tissues of animals infected with the fixed rabies virus. Following are the modified forms.

1. **Semple Vaccine:** Vaccine developed by semple (1911). It is a 5% suspension of **sheep brain infected with fixed virus**

TN_GOVT_XII_Micro_Biology_CH10.indd 220 04-03-2019 12:27:09